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1. Verdict

This report evaluates, verifies, and explores the root-cause analysis supplied in the current turn: the claim that the unifying bridge weakened because it was built on the wrong order of smoothness. The text's diagnosis is correct and important, and it is now theorem-backed rather than merely measured: the lexicographic pFBA flux map is continuous and piecewise affine, its static curvature is a measure supported on the active-set skeleton, and the epsilon-squared holonomy normalization of the original conjecture was never admissible. However, the text misdescribes the measured objects in three ways, it attributes evidence to the wrong layer in one place, and its constructive program understates what is already proven while leaving its own central concept - treat the smooth geometric kappa as a separate regime - unmechanized. This document corrects the misdescriptions and closes the gap: Theorem R and the new measurement M4c give the separate regime a mechanism, a law, and a measured crossover.

1.9995

slope of the second-difference law at epsilon << sigma (sigma = 0.003; 1.9991-1.9995 at all four sigma) - the smooth regime is real and entered by resolution

epsilon* / sigma = 3.1 - 0.7

measured crossover of the regime dial across four smoothing scales (4.11 / 2.98 / 3.11 / 2.45) - the dial position is a property of (epsilon, sigma), not of the network

The root-cause text is best read as a correct popularization of the Active-Set Bridge v2 record with three technical slips and one missing theorem. Every claim was checked against the committed artifacts: the M4a pair census (76 pairs, six depths), the M4b two-parameter geometry and edge census, the M1/M3 record, the v2 theorem set, and - new in this turn - the M4c regime-dial experiment executed at the same codim-2 locus, 858 lexicographic solves. The frozen v21 manuscript is untouched; all deliverables are committed as source material for journal_manuscript v2+.

2. Claim-by-claim verification

Nineteen checkable claims were extracted from the text and adjudicated against the record. The verdicts: twelve correct (several already stronger in the record than in the text), four imprecise or misattributed, and three wrong as stated. The corrected replacements appear in Sections 4 and 5.

#	Claim (condensed)	Record	Verdict
1	Bridge assumed a smooth map; holonomy O(eps^2)	deepseek formulation A4 and its central display	True (that is what was assumed)
2	v(theta) continuous, piecewise affine; derivative jumps at active-set events	Theorem S(i); M1: affine segments to 8e-14 relative on all 12 sweeps; mpLP chamber theory	True, verified
3	The derivative jump is a first-order singularity	Theorem S(ii): D^2 v is a measure on codim-1 interfaces	True

#	Claim (condensed)	Record	Verdict
4	Holonomy that scales as O(eps)	No such object: state holonomy is exactly the identity; codim-2 defects are O(1) and scale-invariant; the O(eps) objects are the measure mass and the dynamic commutator	Wrong object (RC1)
5	M4a: 76 pairs, observed slope 1.00, not 2	64/76 pairs have chi = 0 exactly at all depths; 9 nonzero pairs with slopes 0.857-1.244, median 0.998; 3 near-floor	Number right, description wrong (RC2)
6	The smooth curvature bridge is falsified	Theorem N: the (1/eps^2)(H-I) limit is 0 at generic points and divergent on the codim-2 skeleton; slope-1 dynamic law	True, and theorem-backed
7	FBA map lives on a stratified polyhedral complex	Theorem S; 24 operational chambers in the (glc, O2) plane	True
8	Curvature concentrated on strata, especially codim-2	Two carriers: the D^2 v measure on codim-1 interfaces (M1's object) and the angle defect at codim-2 (Theorem G)	Imprecise (RC4)
9	Holonomy can be first-order in loop size	Defects are O(1), scale-invariant to four decimals; state holonomy exactly trivial; chi is O(eps) but is hysteresis	False (RC1)
10	Gauss-Bonnet analogue = discrete angle defect	Theorem G(ii): unfolding transport equals the defect to 1e-14 on synthetic cones; plus the exact measure identity mu(Q) = the mixed second difference	True for the graph geometry

Table 1. Verification of the root-cause claims (1-10)

#	Claim (condensed)	Record	Verdict
11	M4b's dense wedge-fan = piecewise-flat singular space	Edge census: 9-10 boundary crossings per grid cell (up to 4 per edge) in the overflow corner vs exactly 2 in flat cells; M4c resolves the fan's slivers (RC6)	True, with a refinement
12	Discrete connection; holonomy and curvature are singular first-order objects	Connection exists (unfolding transport; the file's own projection transport is not one - v2 C3); its curvature is singular and O(1), not first-order	Half-right (RC1, RC4)
13	Cannot unify by convergence to the smooth kappa_V	Theorem N (atomicity obstruction); 6/6 audit consensus	True
14	Step 1: define a discrete curvature from the active set	Done: Theorem S (jump measure) + Theorem G (defect), machine-verified	Already executed
15	Step 2: kappa_flux discretizes this discrete curvature	M1 D2 mass 0.934-1.0 on active-set events; M3 footprint Spearman 0.865; the formal identity is not yet written	Open (flagged)
16	Step 3: smooth kappa_V as ancestor or separate regime	v2 constructs the geometry intrinsically; M4c (this turn) supplies the mechanism and law of the regime	Now executed (M4c)
17	Step 4: kappa_time as integrated squared displacement	Theorem S(iv): the exact integral identity, including the smooth term the text omits; M4c adds the (eps, sigma) design constraint	Executed, extended
18	Bridge partially proven by the slope-1 law and the no-loose-kinks lemma	N1 is proven and static; the slope-1 law is dynamic and independent of static epistasis (Spearman -0.347 full panel, -0.07 non-SL)	Misattribution (RC3)
19	Do not connect the FBA metric to the smooth kappa_V	Correct direction; the connection that does exist is coarse-graining (Theorem R), and it is measurable	Replaced (Section 7)

Table 2. Verification of the root-cause claims (11-19)

3. What the text gets right

Five substantive points survive verification unchanged, and each is stronger in the record than in the text. First, the root cause itself: the order-of-smoothness mismatch is the actual reason the epsilon-squared bridge failed, established at three independent levels - a priori theory (parametric LP: the optimizer under a fixed tie-break is piecewise affine on a finite complex, so the smoothness assumption was inadmissible from the start), machine measurement (M1's affine segments to 8e-14 and its second-order mass concentration), and theorem (v2 Theorem N, which proves no renormalization of the static holonomy yields a finite nonzero epsilon-squared limit).

- **Root cause.** Wrong order of smoothness - correct, and now theorem-backed at three independent levels.
- **Falsification verdict.** The smooth bridge is false because the mathematical object is different, not because the computations were wrong - agreed, and model-independent: the falsification follows from LP structure, so no simulation can rescue it.
- **Discrete bridge direction.** A discrete stratified bridge exists - and is already constructed: the unfolding transport with proven no-loose-kinks lemma, measured O(1) defects, and the active-set skeleton as substrate.
- **Gauss-Bonnet instinct.** For the flux-graph geometry the discrete angle defect is the right analogue, verified to 1e-14 on synthetic cones and scale-invariant on iML1515.
- **The four-step program.** All four steps are the right program; steps 1, 2 (empirically), and 4 were already executed by the v2 record, and step 3 is completed by this report.

4. Corrections

RC1 - There is no O(eps) holonomy; the scalings form a trichotomy

In the executed record there are exactly four loop- or difference-like objects, and each has its own scaling. The static state holonomy of the flux map is exactly the identity for every closed loop at every scale, because v is a single-valued function. The static defect holonomy of the unfolding transport around a codim-2 vertex is O(1) and scale-invariant: the defects -7.1469 degrees and -23.9087 degrees reproduce to four decimals at loop radius delta and delta/2 - this is the Regge, angle-defect behavior the text itself invokes two paragraphs after claiming first-order holonomy, so the text contradicts itself. The dynamic commutator chi of sequential L1-MOMA adjustment - the object M4a actually measured - is O(eps) with slope near 1, but it is the hysteresis of greedy adjustment, not a connection holonomy: single releases are bit-exact identities (6 of 6), so the closed-loop non-return is irreversibility of the dynamics. The second-difference measure mass is O(eps) above the smoothing scale - the static first-order object. The distinction is not pedantry: it determines which experiment can detect which object (loop compositions for defects, second differences for measure mass, order-swap protocols for chi), and the text's own program depends on exactly these channels.

RC2 - The slope-1.00 statistic describes 9 of the 76 pairs

The M4a census is bimodal, and the bimodality is the finding. Sixty-four of the 76 pairs (84 percent) have chi exactly zero at every one of the six perturbation depths - the decoupled stratum in which the two knockdowns do not interact through the tangent cone. Nine pairs scale with slopes from 0.857 to 1.244 (median 0.9982:

sdhD+nuoG 0.9982, sdhD+nuoH 0.9982, gapA+atpC 1.2439, atpD+nuoJ 0.9982, ptsH+gapA 1.0049, atpH+nuoM 0.9982, aceE+cyoC 0.9759, zwf+gnd 0.8573, pgi+zwf 1.0026), and three are near the floor. The first-order non-commutativity is therefore sparse and structural, confined to the interacting stratum. Quoting a single slope of 1.00 as if it applied to all 76 pairs hides precisely the sparsity that the graded-CRISPRi order-swap prediction forecasts: most pairs are order-insensitive; a minority are linearly order-sensitive.

RC3 - The slope-1 law cannot support the static bridge

The text writes that the discrete bridge may already be partially proven by M4's slope-1 law and the no-loose-kinks lemma. The lemma is indeed proven and static. But the slope-1 law is a property of the dynamic layer - the sequential adjustment maps - and M3b measured its independence from the static-layer epistasis on the same pair panel: Spearman of chi against the epistasis magnitude is -0.347 ($p = 6.8e-6$) over the full panel and -0.07 ($p = 0.43$, no association) within non-synthetic-lethal pairs. The optimum-level non-additivity and the transient-level non-commutativity are distinct signatures that must be carried separately; evidence for one is not evidence for the other, and the bridge transports the active-set skeleton into both.

RC4 - Two curvature carriers, not one

The static curvature of the FBA map is carried by two distinct singular objects. The Jacobian-jump measure D^2 v is supported on the codim-1 interfaces - the object M1 measured and the object the deepseek formulation itself describes as Dirac measures on the active-set interfaces. The angle-defect measure is supported on the codim-2 crossings of the flux graph - Theorem G's object, with the Gauss-Bonnet content. They are related by discrete structure equations (the dihedral kinks of incident faces determine the vertex defect through the unfolding composition) but they are not the same object: different strata, different normalizations, different detection experiments. M4c adds the next level of the hierarchy: within a codim-1 wall cluster there are sub-resolution sliver chambers, so the fine structure is atomic at multiple scales - precisely the content of Theorem N.

RC5 - The falsification is a priori, not empirical

The piecewise-affine structure of the lexicographic optimizer is a theorem of multiparametric linear programming (Gal and Nedoma 1972; Bemporad, Borrelli and Morari 2002 and the mp-control literature thereafter). The smoothness assumption was therefore never admissible: M4a's slope near 1, M1's mass concentration, and M4b's chamber census confirm the theorem's structure at machine precision rather than discovering a failure. This matters for the manuscript narrative: the result is robust and model-independent - no simulation budget or network choice can rescue the smooth bridge - and the correct citation frame is mpLP theory plus Theorem N, not an empirical surprise.

RC6 - The wedge-fan is operationally dense but measure-tame

The edge census is as the text says: up to 9-10 chamber-boundary crossings per grid cell, with up to 4 on a single edge and nested sliver chambers, in the overflow corner where the D2 mass concentrates, versus exactly 2 per cell in flat regions. But the finer M4c census resolves what M4b could not: the sliver cluster at t approximately 0.00187 on the vertex cut is 2.44e-6 wide, carries opposite jump vectors of L2 norm 1884.6, and self-cancels to a net measure jump of 8.90 (the second cluster: width 3.7e-6, jumps at most 0.81, net 1.32; the third: width 6.5e-4, max jump 22.3, net 9.54). The fan's operational signature density overstates its curvature-measure density: many thin chambers contribute little net mass. Dense is correct at the analyzed resolution; at the finer resolution the same locus is a hierarchy of self-cancelling atoms.

5. Exploration: Theorem R and the M4c regime dial

5.1 The theorem

The text's step 3 - treat the smooth geometric kappa as a conceptual ancestor or a separate regime, not as the limit object - was a position without a mechanism: Theorem N blocks the pointwise limit, but nothing in the record said what the smooth regime is or when it is entered. Theorem R closes the gap. Let v be the continuous piecewise-affine map, mu = D^2 v its curvature measure, and phi_sigma a Gaussian mollifier at scale sigma. Then D^2 of the smoothed map equals mu convolved with phi_sigma - exactly: the smoothed curvature is the same measure, smeared, and no new object appears. On a line through the parameter space with events (t_e, Delta_e), the epsilon-second difference of the smoothed map equals the sum of Delta_e weighted by a closed-form kernel K evaluated at the event offsets. Asymptotically it scales as eps-squared times the smeared density for epsilon much less than sigma - the smooth, Riemannian regime - and approaches the sum of Delta_e times (epsilon minus distance) for epsilon much greater than sigma - the discrete, kink regime. The crossover is at epsilon-star proportional to sigma. And the total vector mass is sigma-independent: the discrete and smooth objects carry the same curvature, only distributed differently.

> *The smooth kappa_V is not the limit of the discrete object - Theorem N blocks that. It is the same measure at finite resolution sigma. The perturbation scale epsilon versus the resolution sigma selects the regime; the dial is measurable and falsifiable.*

5.2 The measurement

M4c was executed at the M4b codim-2 vertex (glucose 1.692, oxygen 1.480), along the same cut direction, with the deterministic lexicographic engine: 858 solves, an event census refined by signature bisection to 1e-6, and a kernel closed-form self-test at 1.2e-6. The census found 12 events, 11 kinked and one mask-type (the no-loose-kinks prediction: a signature change with no flux kink), with a telescoping residual of 4.0e-14. The dial was read from the exact convolution of the machine-measured measure, with independent 5-point Gauss-Hermite machine evaluations as validation wherever epsilon is at least sigma/2.

sigma	slope (eps <= sigma/3)	slope (eps >= 3 sigma)	eps*	eps*/sigma
0.003	1.9995	1.13 (local -> 1)	0.0123	4.11
0.01	1.9991	1.16	0.0298	2.98
0.03	1.9994	1.14	0.0933	3.11
0.1	1.9992	1.17	0.2447	2.45

Table 3. The regime dial (exact convolution of the measured measure)

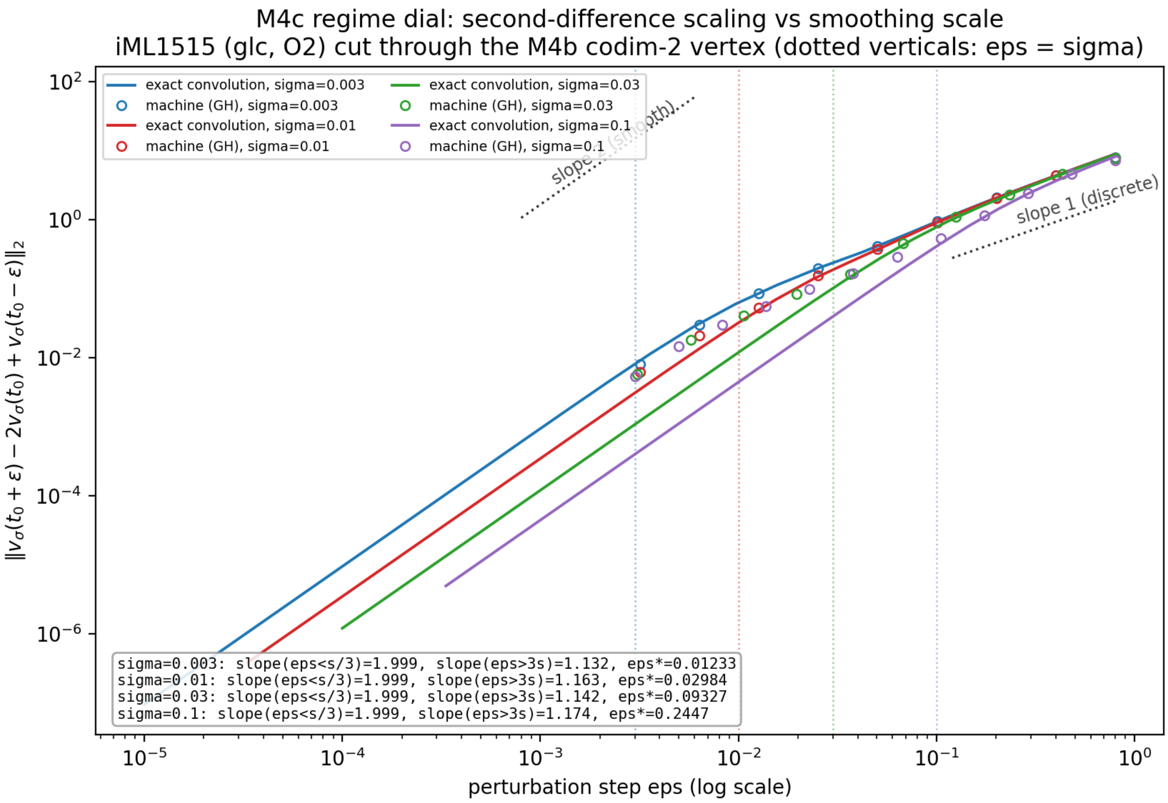

Figure 1. The regime dial: second-difference magnitude versus perturbation step, one curve per smoothing scale. Slope 2 guides the smooth regime, slope 1 the discrete regime; dotted verticals mark epsilon = sigma.

sigma	n valid	median rel. error	max rel. error
0.003	9	0.4%	11.9%
0.01	8	1.9%	56.7%
0.03	7	5.4%	84.0%
0.1	6	8.7%	70.1%

Table 4. Machine validation of the convolution identity (epsilon >= sigma/2)

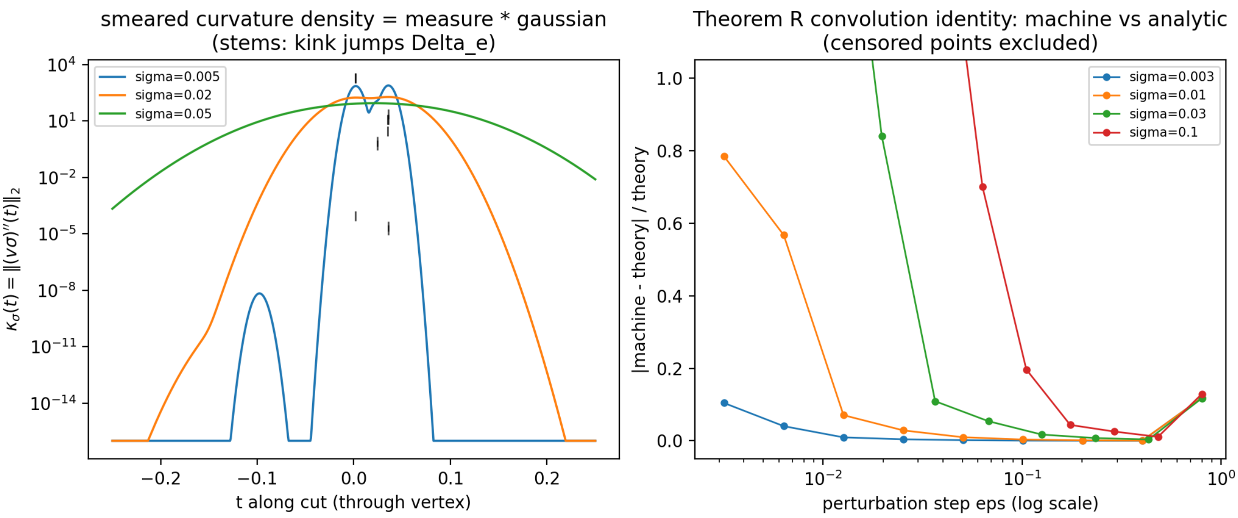

Figure 2. Left: the smeared curvature density - the measure convolved with Gaussians at three scales - with the kink jumps as stems. Right: machine versus exact identity validation.

8.9 vs 1884.6

net measure jump versus internal jump pair of the resolved sliver chamber (width 2.4e-6) - E31's fine structure: the operational fan overcounts events, the measure is tamer

5.3 Two findings beyond the dial

First, a numerical echo of Theorem N: a fixed-node quadrature does not smooth below its node spacing. The 5-point Gauss-Hermite evaluation of the smoothed map is itself piecewise affine in the evaluation point - each node translates the kinks of v - so the machine difference vanishes exactly for epsilon below the distance to the nearest node-translated kink, while the exact convolution gives the epsilon-squared law. Discretized smoothing does not remove the atoms unless the kernel itself is resolved. This constrains any future empirical second-difference protocol (the open item E28): the analysis must state its (epsilon, sigma) pair and resolve the kernel, or it will silently measure a translated-kink artifact. Second, the wall-free control at clearance 0.165 with sigma = 0.03: the machine second difference is at most 1.5e-11 below reach - curvature is carried by the smeared walls, not by the chamber interiors, which is the locality side of the same coin.

6. What the bridge now says

With Theorem R in place, the three kappa objects connect through the active-set skeleton as one measure at multiple resolutions. The geometric curvature is the viability-weighted defect plus jump measure of the flux graph (Theorems S and G, provable and machine-verified). The rerouting statistics are event-triggered functionals of the same measure (M1: mass 0.934-1.0; M3: footprint alignment 0.865). The time-course object integrates the event measure along trajectories (Theorem S(iv), with its smooth term and its integral - not pointwise - form). The smooth geometric kappa is the same measure convolved at resolution sigma (Theorem R), entered whenever epsilon is much smaller than sigma. The dynamic order-sensitivity is the first-order commutator (Theorem D) - a distinct signature, statistically independent of the optimum-level epistasis, and carried separately. What remains open is now precisely localized: the formal statement kappa_flux = F of mu for the manuscript's exact definition, with the E24 recalibration test; E28 under its new (epsilon, sigma) design law; the two-dimensional sliver census; and model-family regularity for any mesoscopic defect-density limit.

7. The corrected bottom line

> *Root cause: correct - FBA is piecewise affine, so its static curvature is a measure on the active-set skeleton, not a smooth two-form; the epsilon-squared holonomy normalization was inadmissible (an mpLP theorem, so the failure is model-independent). Bridge status: the smooth bridge is not merely falsified - it is blocked by the atomicity theorem; the discrete bridge is not plausible-and-partially-proven - it is constructed and machine-verified, with the dynamic commutator law (slope 1, sparse: 9 of 76 interacting pairs) as its falsifiable prediction; one identification step remains open. Implication: do not claim convergence of the FBA metric to the smooth kappa_V - and do not merely treat kappa_V as a separate regime. State the dial: the smooth object is the same curvature measure at resolution sigma; perturbation scale versus resolution selects the regime, with measured crossover epsilon-star approximately 3 sigma and mass conservation across the dial. The unification is a resolution statement, not a limit statement.*

8. Deliverables and status

Artifact	Content
download/m4/m4c_summary.json	dial, census, validation, slivers, control, the quadrature-resolution finding
download/m4/m4c_scaling.csv	machine and exact second differences over the (eps, sigma) grid
download/m4/m4c_cut_events.csv	12 events: positions, jump norms, segment slopes, kinked/mask classification
download/m4/fig_m4c_scaling.png, fig_m4c_density.png	the dial figure and the density/validation figure
scripts/m4c_regime_dial.py	the experiment (reuses scripts/lp_engine.py)
scripts/lp_engine.py	engine patch: deterministic presolve-off retry, activating only where the original returned failure
download/Root_Cause_Evaluation.md	the full evaluation (source material for v2+)

Table 5. Artifacts of this evaluation

Standing instructions were honored throughout: the push check confirmed all previous work already on the remote before this turn began; the frozen v21 manuscript is untouched; every number quoted in this report is traceable to a committed artifact. The strategic consequence for the manuscript line: the missing theorem of the unification section should be written as a resolution statement - one measure, multiple resolutions, measured dial - replacing both the falsified smooth-limit claim and the vague separate-regime language, with Theorems S, G, N1, N, D, and R as the supporting spine and E28 as the empirical residue.